

The Cayley-Hamilton Theorem: A Pillar of Linear Algebra

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Abstract

In this paper, after a short historic introduction, we focus on proving the Cayley-Hamilton theorem and explaining its consequences. This is achieved by first refreshing our memory regarding some fundamental concepts in Linear Algebra, whereafter we prove two distinct theorems, which we will use for the subsequent proof. Finally, the paper explores the implications of the Cayley-Hamilton theorem, offering an understanding of its practical applications and theoretical significance.

1 Introduction

Where it began

On the 10th of December, 1857, an article [?] by Arthur Cayley arrived at the Royal Society, a publisher of scientific journals. This document contained the first steps into Matrix Algebra, nowadays a very powerful mathematical tool, which is used in various applications in for example Control Theory and Machine Learning [?]. One of the key theorems introduced in this article is the now-called Cayley-Hamilton Theorem, so named because a special case of this theorem was proven earlier, in 1853, by William Hamilton [?].

After a concise introduction of this theorem, we will elaborate on some of the prerequisites, give a proof of this famous theorem, and finally, end this paper with some of its consequences for modern mathematics.

The gist

The Cayley-Hamilton Theorem only applies to square matrices, which are matrices whose length and width are equal, e.g.

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_{\text{Square}} \qquad \underbrace{\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix}}_{\text{Non square}}$$

The usefulness of matrices with this characteristic cannot be understated: They represent a set of linear equations that has the same amount of equations as unknowns, thus increasing the likelihood of finding a solution.

Cayley-Hamilton's theorem states that, for any square matrix, if we put this matrix in its own characteristic polynomial, the result will always equal a zero-matrix. What this means in practice, is that, since the characteristic polynomial encodes important information about eigenvectors and their eigenvalues, these properties will be less arduous to find.

Since eigenvectors allow us to diagonalize matrices, more on that later, they severely remove some of the complexities of solving matrix calculations. Consequently since solving matrices is used in any field that utilizes multiple linear equations, and since Linear Algebra was created to solve sets of linear equations easier, it is not stated lightly that this theorem really stands as a pillar on which Linear Algebra thrives.

Now for a concrete example, let

$$A = \begin{pmatrix} 7 & 2 \\ 1 & 5 \end{pmatrix}.$$

Then by definition, the characteristic polynomial, more on this later, of A can be written as

$$f(A) = \det(A - I_2\lambda) = \det \begin{pmatrix} 7-\lambda & 2 \\ 1 & 5-\lambda \end{pmatrix} = \lambda^2 - 12\lambda + 33,$$

where

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then the Cayley-Hamilton Theorem tells us that

$$A^2 - 12A + 33I_2 = 0,$$

which we can easily check,

$$\begin{pmatrix} 7 & 2 \\ 1 & 5 \end{pmatrix}^2 - 12 \begin{pmatrix} 7 & 2 \\ 1 & 5 \end{pmatrix} + \begin{pmatrix} 33 & 0 \\ 0 & 33 \end{pmatrix} = \begin{pmatrix} 51 & 24 \\ 12 & 27 \end{pmatrix} - \begin{pmatrix} 84 & 24 \\ 12 & 60 \end{pmatrix} + \begin{pmatrix} 33 & 0 \\ 0 & 33 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

A formal representation of this theorem is obviously due:

Theorem 1.0.1. (see e.g. [?, p. 317]) *Let T be a linear operator on a finite-dimensional vector space V , and let $f(t)$ be the characteristic polynomial of T . Then $f(T) = 0$, the zero transformation.*

To prove this, we will use a combination of T -Cyclic subspaces and two separate theorems that we will elaborate later, to show that $f(T)(v) = 0$ for all $v \in V$.

2 Prerequisites about Linear Algebra

To understand parts of the proof of the Cayley-Hamilton Theorem, we will have to define and explain some fundamental concepts beforehand.

2.1 Linear Operators

Linear Operators are what their name implies. They represent a mapping, from a space to itself, that is linear in its nature.

Definition 2.1.1. Let V be a vector space, then any mapping $T : V \longrightarrow V$ where $T(av_1 + bv_2) = aT(v_1) + bT(v_2)$ with $a, b \in \mathbb{R}$ and $v_1, v_2 \in V$ can be called a linear operator [?, p. 112].

We define the matrix representation of T relative to an ordered basis β , by the notation $[T]_\beta$.

2.2 Eigenvectors and their eigenvalues

Definition 2.2.1. Let T be a linear operator on a vector space V . A nonzero vector $v \in V$ is called an *eigenvector* of T if there exists a scalar λ such that $T(v) = \lambda v$. The scalar λ is called the *eigenvalue* corresponding to the eigenvector v [?, p. 246].

Eigenvectors, in the realm of linear algebra, are non-zero vectors that maintain their direction under a linear transformation. This is typically represented by the equation $Av = \lambda v$, where A is a matrix representing the linear transformation, v is the eigenvector, and λ is the corresponding eigenvalue, which is the factor by which the eigenvector is stretched. Matrices can be viewed as a representation of linear transformations. These transformations can involve operations like rotation, stretching, or shearing of vectors. Eigenvectors are particularly significant because they provide a static reference point that remains unchanged under any transformation. This makes them invaluable for understanding how linear transformations behave on specific objects. Even if the initial plane undergoes a transformation such as stretching or rotation, its eigenvectors remain fixed, allowing us to easily comprehend the effects of the transformation.

2.3 The Characteristic Polynomial

The Characteristic Polynomial is the first step in getting a greater understanding of matrices in general.

Definition 2.3.1. Let A be a square matrix, then $f(\lambda) = \det(A - \lambda I)$ is its characteristic polynomial [?, p. 248].

We are interested in applying this operation to our matrix A because it gives us a very easy way to figure out its eigenvalues. An example of the characteristic polynomial in practice can be seen in the introduction of this paper.

2.4 T-Invariant subspaces

T -invariant subspaces are subspaces of a vector space that are completely self-contained on transformation. Thus any vector in the subspace, under a certain transformation T will always transform into another vector of that same subspace. More formally:

Definition 2.4.1. Let T be a linear operator on a vector space V . A subspace W of V is called a T -invariant subspace of V if $T(W) \subseteq W$, that is, if $T(v) \in W$ for all $v \in W$ [?, p. 313].

By dividing a vector space, if possible, into T -invariant subspaces, we allow ourselves to study each T -invariant subspace independently. In general, it is easier to study multiple smaller independent vector spaces compared to one larger vector space, thus this makes our job easier.

We can then define

$$W = \text{span}(x, T(x), T^2(x), \dots),$$

to be the T -cyclic subspace of V , generated by x .

3 Proof of the Cayley-Hamilton Theorem

3.1 Linear dependence

Theorem 3.1.1. Let S be a linearly independent subset of a vector space V , and let v be a vector in V that is not in S . Then $S \cup \{v\}$ is linearly dependent if and only if $v \in \text{span}(S)$ [?, p. 39].

The proof of this theorem is out of the scope of this paper.

3.2 Division of the characteristic polynomial by its T-invariant subspace

Theorem 3.2.1. *Let T be a linear operator on a finite-dimensional vector space V , and let W be a T -invariant subspace of V . Then the characteristic polynomial of T_W , that is, the restriction of T to only W divides the characteristic polynomial of T [?, p. 314].*

Proof. We choose an ordered basis $\alpha = \{v_1, v_2, \dots, v_k\}$ of W . Then since W is a subspace of V , we can extend α such that it becomes an ordered basis for V , $\beta = \{v_1, v_2, \dots, v^k, v^{k+1}, \dots, v^n\}$. If we let $A = [T]_\beta$ and $B_1 = [T_W]_\alpha$, then it follows that since W is a T -invariant subspace of V , by definition, $T(v_i) \in W$ for all $v_i \in \alpha$. Thus the first v_1 to v_k bases in β are only used to represent W , and as a result, A can be uniquely represented as a matrix with a zero in one corner:

$$A = \begin{pmatrix} B_1 & B_2 \\ 0 & B_3 \end{pmatrix}.$$

If we then let $f(t)$, $g(t)$ be the respective characteristic polynomial of T and T_W , it is easily derived that

$$\begin{aligned} f(t) &= \det(A - tI_n) = \det \begin{pmatrix} B_1 - tI_k & B_2 \\ 0 & B_3 - tI_{n-k} \end{pmatrix}, \\ &= \det(B_1 - tI_k) \cdot \det(B_3 - tI_{n-k}) = g(t) \cdot \det(B_3 - tI_{n-k}), \end{aligned}$$

and consequently,

$$\frac{f(t)}{g(t)} = \det(B_3 - tI_{n-k}),$$

dividing

$$f(t). \quad \square$$

3.3 T-cyclic subspaces and their bases

Theorem 3.3.1. [?, p. 315] *Let T be a linear operator on a finite-dimensional vector space V , and let W denote the T -cyclic subspace of V generated by a nonzero vector $v \in V$. Let $k = \dim(W)$. Then*

- (a) $T^0(v), T(v), T^2(v), \dots, T^{k-1}(v)$, the collection of multiples of $T(v)$, is a basis for W .
- (b) If $a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = 0$, then the characteristic polynomial of T_W is $f(t) = (-1)^k(a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k)$.

Proof.

- (a) Given a non-zero vector v in a finite-dimensional vector space V , we can form a set

$$\beta = \{v, T(v), T^2(v), \dots, T^{j-1}(v)\},$$

where T is a linear transformation and j is the largest positive integer such that β is linearly independent. We then define Z as the span of β .

Since β is a subset of Z , it is clear that Z is a T -invariant subspace of V . Furthermore if $w \in Z$, then w can be expressed as a linear combination of the vectors in β . Therefore, applying the linear transformation T to w results in a vector that is a linear combination of the vectors in Z , implying that $T(w)$ is also in Z .

We also know that $T^j(v) \in Z$ according to Theorem 3.1.1. Furthermore, $v \in Z$ as it is a linear combination of the vectors in β . Therefore, Z is the smallest T -invariant subspace of V

that contains v , denoted as W . It follows that Z is a subset of W and W is a subset of Z . Hence, Z and W must be equal, and β forms a basis for W . This implies that $\dim(W) = j$. and thus $j = k$.

(b) Now view β , from (a), as an ordered basis for W . Let $a_0, a_1, \dots, a_{k-1}$ be the scalars such that

$$a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = 0.$$

Observe that

$$[T_W]_\beta = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{k-1} \end{pmatrix},$$

which by some induction on k , which is outside the scope of this paper, gives us a characteristic polynomial

$$f(t) = (-1)^k(a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k). \quad \square$$

3.4 Cayley-Hamilton

We now prove the Cayley-Hamilton Theorem 1.0.1.

Proof. We aim to demonstrate that for any vector $v \in V$, the function $f(T)(v) = 0$.

Given that T is a linear transformation, we can set $v = 0$ to show that $f(T)(0) = f(0) = 0$. This outcome is a direct result of the properties of vector spaces, where the zero vector, having no length or direction, remains at the origin regardless of any applied transformations.

Now suppose $v \neq 0$, let's consider W as a T -cyclic subspace produced by v , and assume the dimension of W is k . Then it follows from Theorem 3.3.1(a) that there exist scalars $a_0, a_1, \dots, a_{k-1}$ such that

$$a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = 0.$$

Subsequently, by Theorem 3.3.1(b), the characteristic polynomial of T_W is:

$$g(t) = (-1)^k(a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k).$$

When we combine these two equations, we find that:

$$g(T)(v) = (-1)^k(a_0I + a_1T + \dots + a_{k-1}T^{k-1} + T^k)(v) = 0,$$

This equation indicates that, as per Theorem 3.2.1, given $g(t)$ is a divisor of $f(t)$, there must exist a polynomial $q(t)$ such that $f(t) = q(t)g(t)$ and thus finally

$$f(T)(v) = q(T)g(T)(v) = q(T)(g(T)(v)) = q(T)(0) = 0. \quad \square$$

4 Consequences of the Cayley-Hamilton Theorem

4.1 Powers of matrices

The Cayley-Hamilton theorem establishes a fundamental relationship between a matrix and its powers. This relationship allows for the simplification of expressions involving these powers,

enabling their evaluation without the need to calculate the power of the matrix or any higher powers. Consider an example where A is a matrix

$$A = \begin{pmatrix} 2 & 4 \\ 2 & 1 \end{pmatrix}.$$

According to the Cayley-Hamilton theorem, we have

$$f(t) = (2 - t)(1 - t) - 8 = t^2 - 3t - 6 = 0,$$

it follows that

$$f(A) = A^2 - 3A - 6I = 0,$$

and thus

$$A^2 = 3A + 6I.$$

To calculate A^3 , we multiply A^2 by A :

$$A^3 = A^2 A = (3A + 6I)A = 3A^2 + 6A = 3(3A + 6I) + 6A = 15A + 36I,$$

thus making it very easy for us to find higher powers of matrices in an alternative way [?, p. 318].

4.2 Inverses of matrices

In the same way as the previous argument, we can also easily find the inverse of any square matrix by applying the Cayley-Hamilton Theorem. Again consider the same example, let A be a 2×2 matrix

$$A = \begin{pmatrix} 2 & 4 \\ 2 & 1 \end{pmatrix}.$$

Then

$$f(A) = A^2 - 3A - 6I = 0,$$

and thus

$$6I = A^2 - 3A, \quad 6I = A(A - 3), \quad A^{-1} = \frac{1}{6}(A - 3) = \frac{A}{6} - \frac{1}{2}.$$

It then follows that

$$A^{-1} = \begin{pmatrix} \frac{-1}{6} & \frac{1}{6} \\ \frac{-1}{6} & \frac{1}{3} \end{pmatrix}.$$